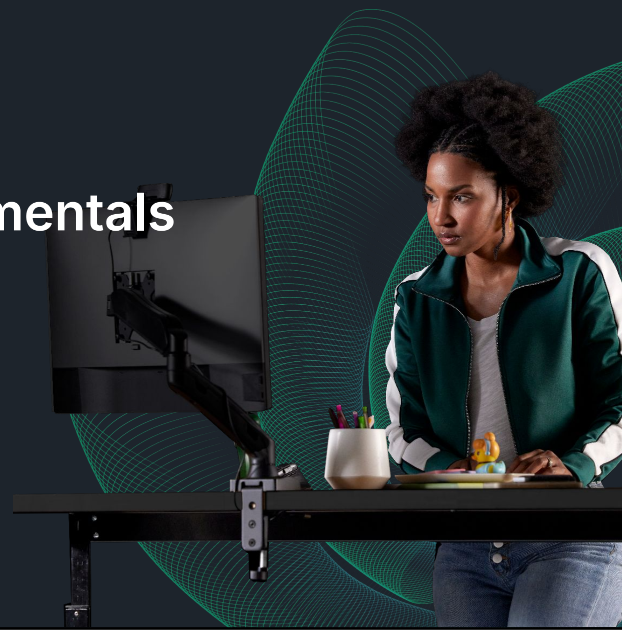
OpenTelemetry Fundamentals

Phil Weber
New Relic University



Agenda

01	What is OpenTelemetry?
02	Instrumentation methods
03	Sampling
04	Collectors
05	OpenTelemetry in New Relic

new relic

What is OpenTelemetry?

Observability Framework

A collection of tools, APIs, and SDKs. It is **not** a backend, but a standard way to generate and capture data.

Vendor-Agnostic

Designed to prevent vendor lock-in. You own your data and can send it to any supported backend.

Three Pillars

Standardizes the collection of the three primary telemetry signals: **Traces**, **Metrics**, and **Logs.**

new relic

Instrumentation Methods

</> Code-Based

Requires modifying the application source code using OpenTelemetry APIs and SDKs.

- **Pro:** Deep, custom insights.
- **Pro:** Full control over data context.
- **Con:** Requires engineering effort.

Zero-Code

Uses agents or libraries to automatically attach to the running application without code changes.

- **Pro:** Fast and easy setup.
- **Pro:** Good for “black box” apps.
- **Con:** Limited to standard signals.

new relic

Sampling

Head-Based Sampling

The decision to sample a trace is made at the **beginning** of the request.

"Do we keep this random trace?"

- Simple and low overhead.
- Risk of missing interesting errors if they are rare.

Tail-Based Sampling

The decision is made **after** the trace has completed.

"Did this trace have an error? If yes, keep it."

- Ensures you capture errors and high-latency traces.
- Higher resource cost (must buffer data).

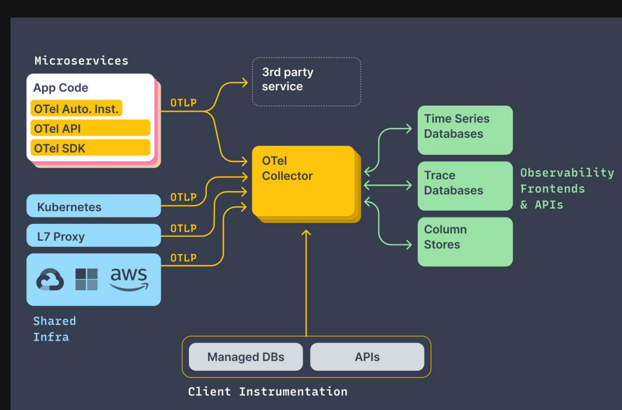
new relic

The OpenTelemetry Collector

A Vendor-Agnostic Proxy

The Collector is a stand-alone service that receives, processes, and exports telemetry data.

- **Receivers:** How data gets in (e.g., OTLP)
- **Processors:** How data is modified (e.g., batching, obfuscation).
- **Exporters:** Where data goes (e.g., New Relic, Prometheus)



new relic

Why Use a Collector?

Offloading

Removes the responsibility of sending data from your application, improving app performance.

Data Control

Centralized place to filter sensitive data (PII), batch requests, and handle retries.

Routing

Send the same telemetry data to multiple backends simultaneously (e.g., New Relic + Prometheus) without changing code.

new relic

OpenTelemetry vs. New Relic Agents

Feature	OpenTelemetry	New Relic Agent
Philosophy	Vendor-neutral, open standard	Proprietary, platform-optimized
Flexibility	High (Any backend, custom processing)	Low (Tied to New Relic ecosystem)
Setup Effort	Moderate (configuration required)	Low ("out of the box" experience)
Ecosystem	Community-driven, vast library support	Curated, vendor-supported

new relic

Using OpenTelemetry with New Relic

Native OTLP Support

New Relic natively supports the OpenTelemetry Protocol (OTLP). You do not need a proprietary agent to send data.

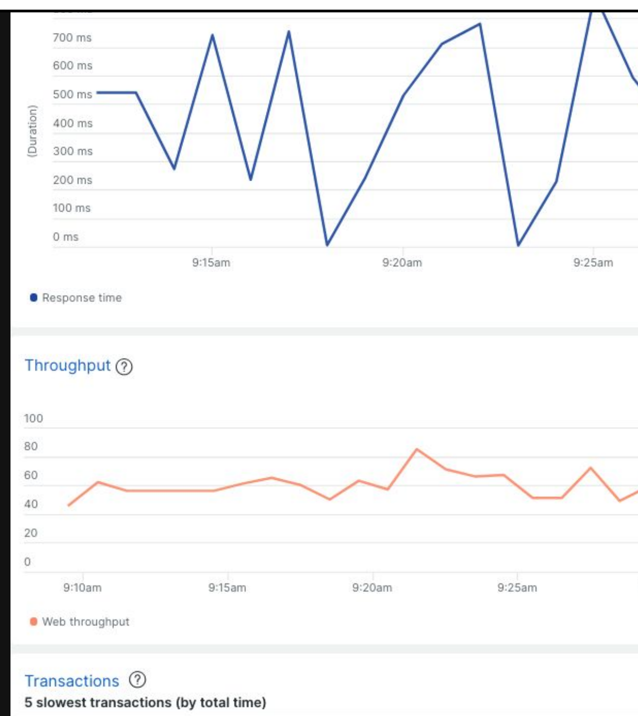
- Simply point your OTel Collector or SDK exporter to the New Relic OTLP endpoint.
- **Endpoint:** otlp.nr-data.net:4317
- Requires only your License Key for authentication.

new relic

Unified Experience

New Relic's *APM Convergence* normalizes OpenTelemetry data to look and feel like data from New Relic agents.

This provides a unified view where you can see services instrumented with both methods side-by-side in the same service maps and dashboards.



Learn more

- OpenTelemetry documentation
- New Relic OTel documentation
- New Relic OTel examples (Github)

new relic